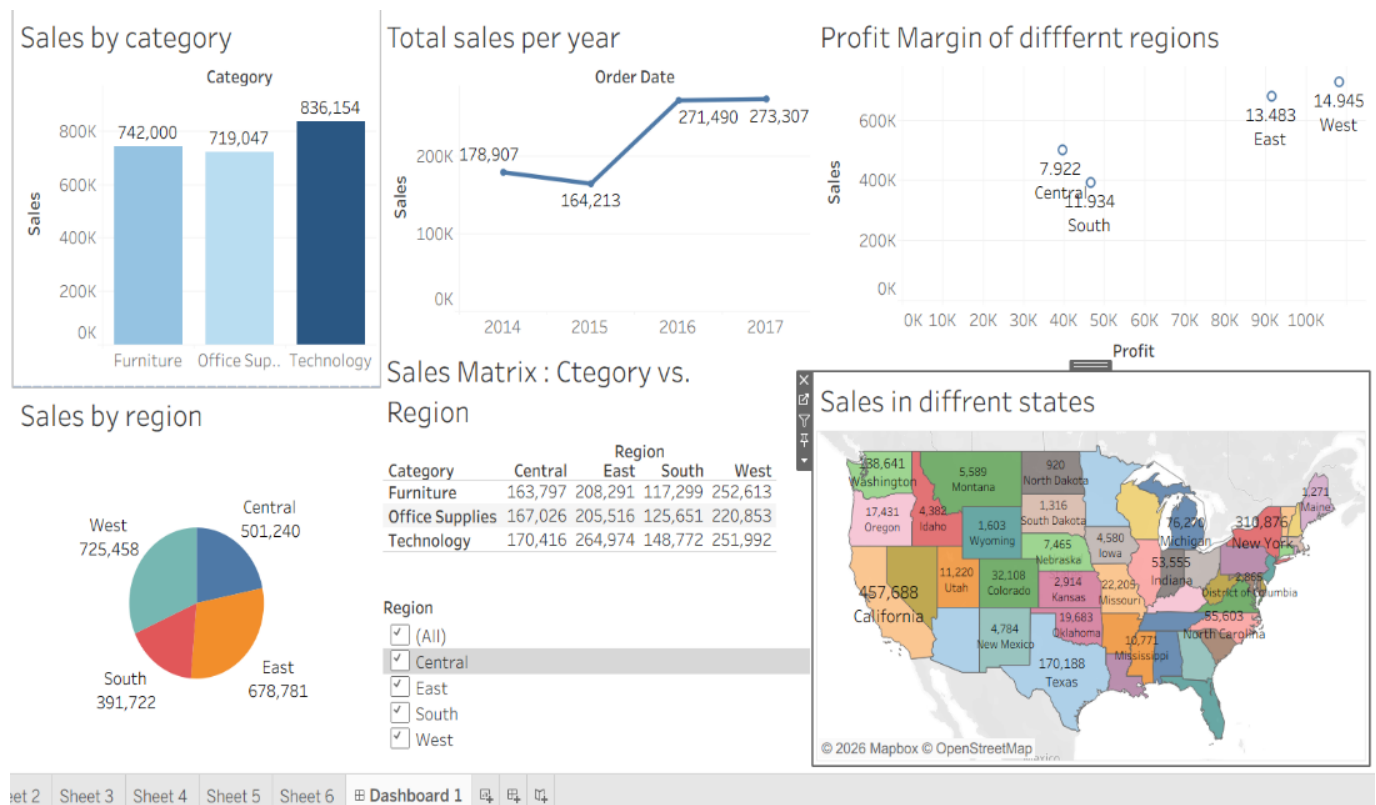


Executive Summary

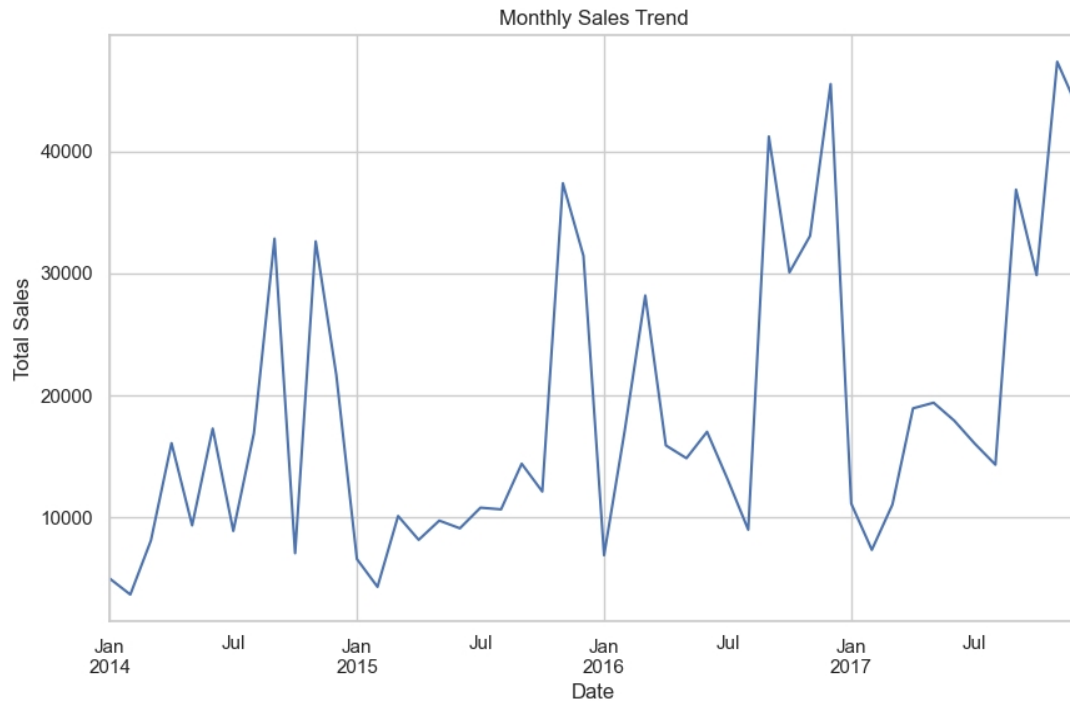
This report summarizes a 30-day data analytics project conducted on the Superstore Sales dataset (2014–2017, ~9,994 orders), covering exploratory analysis, SQL-based extraction, interactive dashboarding, and advanced statistical modeling. The analysis reveals steady year-over-year revenue growth, a small set of high-value customers driving disproportionate profit, and clear opportunities in regional profitability and customer retention. Below are the top insights and actionable recommendations for the business.

Dashboard

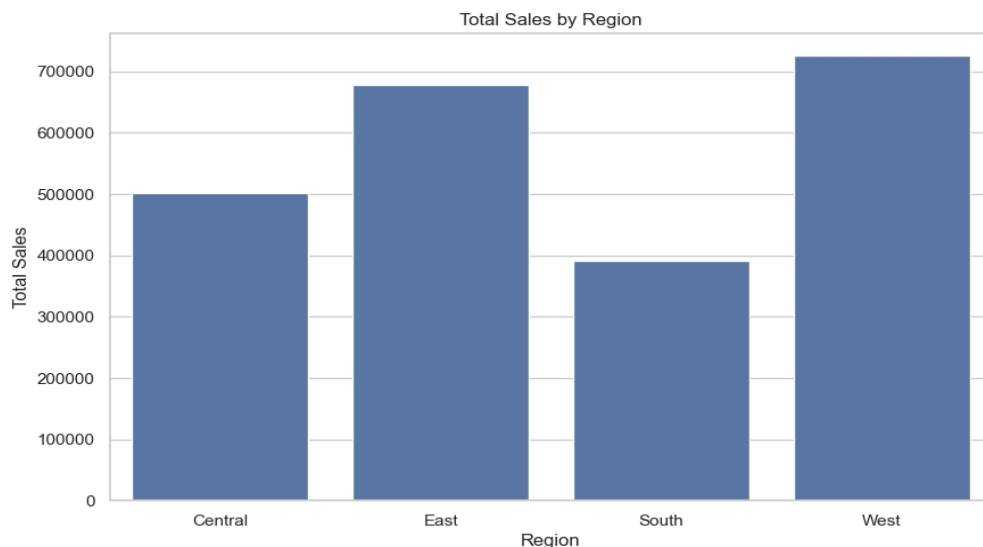


Top 5 Insights

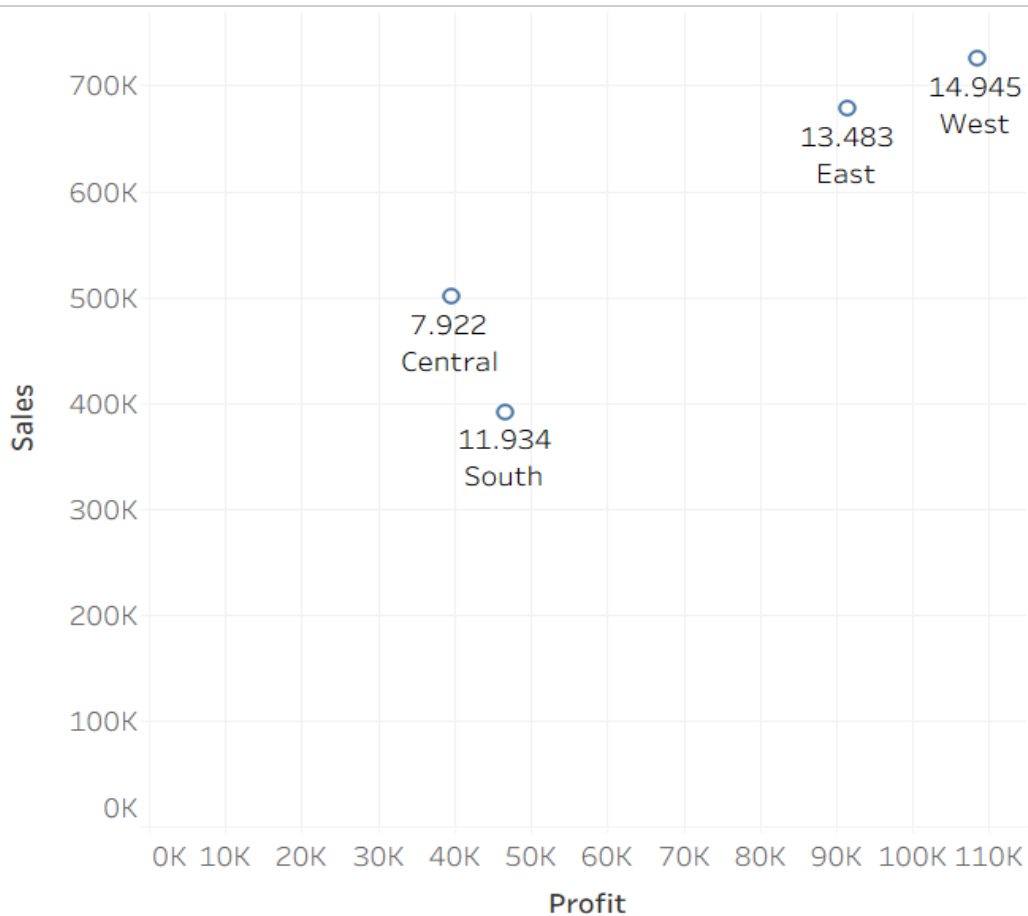
- 1. Revenue grew consistently year-over-year, nearly doubling from 2014 to 2017** Total sales rose from ₹178,907 (2014) to ₹273,307 (2017) — a strong upward trend, though 2015 saw a dip to ₹164,213 before recovering sharply in 2016–2017.



- 2. Technology drives the most revenue, but profitability varies sharply by region** Technology leads all categories in total sales (₹836,154), ahead of Furniture (₹742,000) and Office Supplies (₹719,047). However, regional profit margins tell a different story — the West region leads in profitability (14.95%), while Central lags significantly behind (7.92%), despite both selling similar product mixes.



Profit Margin % of difffernt regions



3. A small segment of customers accounts for a disproportionate share of value RFM-based clustering identified a high-value "VIP" segment representing customers with the highest average profit per order (₹170.67) — roughly 8–15x higher than any other segment — despite not being the most frequent buyers. This group is small but critical to overall profitability.

4. A significant portion of the customer base is at risk of churn One customer segment showed an average recency of 851 days since last purchase — over two years of inactivity — combined with the lowest purchase frequency of any group. This lapsed segment represents a meaningful reactivation opportunity.

	recency	frequency	monetary	avg_profit
cluster				
0	323.49	6.09	9763.65	170.67
1	851.13	3.14	2072.87	20.95
2	166.06	3.99	2089.29	11.62
3	141.66	10.67	4013.39	25.90

5. Sales and profit are only weakly linked — high sales don't guarantee high profit Statistical testing confirmed no significant difference in profitability between Consumer and Corporate segments ($p = 0.39$), and no significant association between region and product category purchased ($p = 0.72$). A predictive model built to forecast profit from order characteristics performed poorly ($R^2 = -0.64$), indicating profit is driven by more complex factors — likely discount depth — that a simple linear approach can't capture.

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R^2:  -0.6440
MAE:  67.7047
RMSE: 282.3307
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3 Business Recommendations

1. Prioritize regional profit optimization over pure sales growth Since Central region sales volumes are healthy but margins are the weakest, investigate discount policy and cost structure specifically in that region rather than pushing for more volume — the data shows volume alone isn't translating to profit there.

2. Launch a targeted win-back campaign for the lapsed customer segment With average recency exceeding 2 years for a meaningful customer segment, a time-sensitive reactivation campaign (personalized discounts, "we miss you" outreach) could recover revenue before these customers are permanently lost.

3. Investigate the discount-profit relationship before broadening discount programs Given the weak/negative link between sales and predictable profit, run a focused analysis on discount thresholds specifically — the current data suggests heavy discounting may be quietly eroding margins on a subset of orders, which merits review before any broader promotional expansion.
